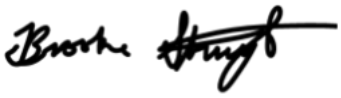
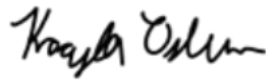
Document Title: BME Design Specifications, Constraints, and Hazards		
Document #: DHF 002	Revision #: 02	Previous Doc/Rev #: DH001-01

CHANGE HISTORY

Rev.	Date	Detailed change description (include rationale for change)	Affected Documents	Supporting Documents
01	08/27/24	Initial Release	DH01	EN-1
02	10/20/2026	2nd Revision	DH02	EN-1, DH01, EN-2

DOCUMENT REVIEW AND APPROVAL

Name	Title	Contributed Sections	Signature/Date
Brooke Struyk	Project Lead and Design Engineering	Code of Ethics & Study Approval Committees, Ethical Dimensions and Analysis	 10/20/2026
Kaayla Osborne	Client POC and Electronic Engineering	Environmental & economic factors, and design inputs	 10/20/2026
Rachel Mensah	Testing and Validation Engineering	Purpose, document definitions, global, cultural, and social external factors, and Preliminary Hazard Analysis (PHA)	 10/20/2026
Sarah Krishnamoorthy	Software and Systems Engineering	Document acronyms, External factors 3 – public health safety, welfare, and House of Quality	 10/20/2026

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Purpose

This document outlines the design inputs, including engineering requirements, specifications and potential hazards associated with the waste anesthetic gas active scavenging system being developed in collaboration with KidsOR and Public Invention. This scavenging system will reduce healthcare personnel exposure to waste anesthetic gas in low-resource operating rooms. As the second document in the Design History File (DHF) series, this document is intended to communicate important design considerations and requirements. This document evaluates external factors including global, cultural, social, environmental, economic, public health, safety, and welfare considerations, that may influence the design and implementation of the system. Additionally, it addresses ethical and professional responsibilities through engineering code of ethics, study approval committees, and ethical considerations. The document also establishes design inputs, identifies preliminary hazards, and uses a House of Quality to connect user needs with engineering specifications. Overall, this document serves as a foundation for guiding design decisions and ensuring the creation of a safe and effective scavenging system.

Definitions

Document Definitions

Table 002-1 Document Definitions

Term	Definition
Design History File	A compilation of records containing the complete design history of a finished device or service.
User Need	A clear statement of what a user requires from a product or system to complete a task or solve a problem.
Design Input	The design requirements relating to a device that are relevant and appropriate for the device’s intended use and user needs.
Verification	Confirmation by examination and provision of objective evidence that specified requirements have been fulfilled.
Validation	Establishing objective evidence that system specifications conform to user needs and intended uses.
Hazard	Potential harm, injury, illness, or damage caused by a source or process.
Foreseeable sequence of events	A series of steps or situations that can be anticipated and lead to a hazardous situation or potential harm.
Hazardous situation	Any circumstance or event where property, people, or the environment are exposed to potential sources of harm.

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Harm	Physical or mental injury, damage or pain inflicted upon a person, object or system.
Risk	Possibility of an undesirable outcome, loss, injury, or danger.
Residual Risk	Remaining level of risk after safeguards, mitigation measures and controls have been applied.

Document Acronyms

Table 002-2 Document Acronyms

Acronym	Description
DHF	Design History File
ISO	International Organization for Standardization
IEC	International Electrotechnical Commission
ASTM	American Society for Testing and Materials
ANSI	American National Standards Institute
WAG	Waste anesthetic gas, gas that escapes patient breathing circuit/current scavenging system
AGSS	Anesthetic gas scavenging system
OR	Operating Roon
BMET	Biomedical Engineering Technician
PPE	Personal Protective Equipment
UN	User Need
DI	Design Input

Design Constraints

External Factors 1 - Global, Cultural, and Social

This device is being developed to reduce healthcare personnel exposure to waste anesthetic gases (WAGs) in low-resource operating rooms internationally. Since KidsOR supports surgical care in different countries, global differences in healthcare infrastructure, resources, and regulations must be

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considered. Operating rooms in lower-resource areas may have limited access to reliable electricity, replacement parts, existing gas disposal or filtering systems, and medical equipment. These limitations could impact the installation, operation, and maintenance of this device in the long term. The device design should prioritize affordability, durability, portability, and ease of maintenance. This may require using locally available materials and components, alternative power sources such as reusable batteries, and incorporating recyclable and replaceable parts. These decisions could help ensure that healthcare personnel are more receptive of the device because it “easily” blends into their specific environment.

Another major global consideration is the differences in medical device regulations, engineering standards, and safety requirements. For example, ISO 7396-2 provides requirements for anesthetic gas scavenging systems including the performance, installation, and safety, but the question then becomes if this standard (among others) is applicable internationally; there may be country specific regulations and standards that our device must abide by [1]. This introduces the user needs of accessibility, affordability, reliable operation, and ease of use while the design inputs include low power requirements, replaceable components, and compatibility with locally available hospital equipment.

Cultural factors may influence how healthcare personnel and patients interact with the device when introduced into operating rooms in different countries. Medical practices, training expectations, familiarity with technology, communication styles, and languages vary across operating rooms. Some operating rooms have personnel who are already familiar with active scavenging systems while others may have limited knowledge and experience using these systems. Language differences can also create challenges when understanding operating instructions, safety warnings, equipment labels, and maintenance procedures. If instructions are only in English, some healthcare workers may have difficulty using the system correctly if they primarily speak other languages. Taking this into account, the scavenging system should include clear visual instructions, recognizable symbols, and numbered steps that make the operating process easy to understand. Training materials might be able to be translated based upon the country the device is in. These factors emphasize the user-friendly operating procedures and clear communication of safety information.

Social factors also extend to healthcare personnel's comfort and acceptance of this device. Since the system may operate near operating room staff, its physical appearance, positioning, and noise level may influence how healthcare personnel interact with it. A device that is bulky, noisy, or difficult to position could interfere with existing workflows and discourage consistent use. Therefore, the system should be designed to operate and integrate into existing operating rooms. This consideration focuses on developing a user-friendly system that healthcare personnel can consistently use without disruption to their current procedures and techniques.

External Factors 2 - Environmental and Economic

Volatile waste anesthetic gases like sevoflurane, isoflurane, and desflurane are greenhouse gases that damage our atmosphere if released into the environment in their natural state [2]. One of our team’s main user needs is to remove WAG from the open air in the operating room, but a large thing to consider is where these gases will go once removed. Because we don’t want to contribute to pollution of greenhouse gases in the environment, the team is developing a design that uses adsorption to collect the WAG before it is released to the environment. This introduces design inputs that focus on

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the method of collection, the type of material being used to adsorb, and what it looks like when that material is at the end of its life cycle.

This brings us to the next environmental factor of our product lifecycle. Our potential design includes a method to actively remove WAGs from the OR, the adsorption material, tubing the gas will be moved through, a power source, different types of sensors/displays, and a pressure relief valve. All these components have their purpose withing our system along with their own life cycle. There are parts of the system that are meant to be permanent, such as the pressure relief valve, the tubing, and the sensors/displays, but they may still need replacements. In these cases, it is likely that the old parts will be placed in a landfill and unlikely to be repurposed, but that is why we need to choose materials that are durable so that they last a long time which will be reflected in our design inputs. Other components of the system, like the adsorption material, are expected to be routinely replaced due to more active use. In this case we should still maximize the number of times it will be used; however, we should be more considerate of how it will be disposed of or if it can be recycled.

The environment our device will be used in is also a huge factor to think about when we are designing our product. The operating rooms that the product will occupy are all spread across Africa. This means we must consider how our device responds to high temperatures and humid climates. Because we are collecting WAG through tubing in a humid environment, there is potential for the gas to become condensate and mold the tubing. Finding solutions to detect the mold or methods to prevent mold is important as to not spread mold spores in the operating room, which is dangerous to the medical personnel and extremely dangerous to the patient being operated on since they will have a weakened immune system.

A large economic factor that we must consider is how much energy our product will consume. In Africa, there is not always guaranteed power because it is intermittently shut off. Hospitals then rely on solar power or other backup power sources to operate [3]. This backup power is used to keep important machines running like ventilators and imaging machines, so our product should not rely on this as a backup power source. This means our design should integrate a backup power source within the product. Implementing a modular power source in our product can make it more expensive to manufacture because we will have to pay for extra storage space and most likely batteries. That extra cost will make our product more expensive but will improve the quality of use since it will still run in a power outage.

Higher quality tubing is another factor that can initially drive-up costs. The current tubing being used to passively move the WAGs is a corrugate tubing made of thin plastic that breaks easily which contributes to the WAG being released into the open environment. This tubing is cheap but in the long term will cost more money to use because of having to constantly replace it. If we use a material like PVC, there will be a higher cost to buy the first set of materials, but it will last longer and has more benefits. A rigid structure like PVC being used to move the WAG will make cleaning easier and reduce the number of kinks in the system from being stepped on since it is more durable than corrugated tubing. PVC is also extremely accessible in Africa, so there is no concern about international import costs.

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External Factors 3 - Public Health, Safety, and Welfare

There are many external factors in this project and KidsOR's mission itself. The primary concern of KidsOR is public health. KidsOR's mission is to improve access to pediatric surgery for patients in low-resource settings. However, because of limitations in the retrofitted ORs, appropriate WAG routing is difficult and can make disposal difficult, resulting in WAG release into the OR, exposing staff. Two major limitations are room and hospital layout as well as centralized hospital vacuum infrastructure availability. Given most of the buildings lack a centralized vacuum or gas disposal system, the majority rely on passive system via a plastic breathing-circuit tubing routed away from the OR. The gas may be routed outside or in other inside common areas, such as waiting rooms, so even if gas is routed away from staff in the OR, other medical personnel, staff, and patients may be exposed to WAGs. Additionally, if a patient's breathing is not strong enough, the potential for WAG to escape or stagnation in the OR increases. This inconsistent and unreliable method of disposal clearly illustrates a need for an active scavenging system that effectively removes the gas and does not require reliance on a centralized hospital system. These support our user needs that the device actively removes WAG from the OR (UN-3), operates independently of centralized infrastructure, and maintains adequate WAG flow during usage. The design input requires the device to achieve at least a 90% reduction in WAG concentration (DI-3), whilst the pressure and scavenging flow rate are also evaluated.

Another crucial consideration is the public health concern of the environmental impact of improper WAG disposal. The current system typically relies on a passive system that simply moves the gases out of the OR, instead of treating or capturing the gas. The team is considering a design that not only more actively moves gas out of the OR but also employs a low-cost treatment prior to exhausting the gas. The team has considered adsorption via materials like activated carbon, but the cost, replacement, and availability of the materials in low-resource environments. This relates to the user needs of a low-cost device whose replacement parts are readily available. Based on the team’s Design Control Matrix, the solution must use inexpensive materials (DI-13) and accessible/reproducible components (DI-12). An alternative currently being considered by the team is a packed scrubber utilizing inexpensive materials. Availability of replacement materials is an important user need as well, because otherwise there will be little benefit for KidsOR to adopt the solution. The solution would also require a way of showing when the treatment medium requires replacement. This is part of the user need requiring accurate WAG detection (UN-5), easily understandable output (UN-6), and a clearly labelled alarm system (UN-15). The design inputs for these user needs require a WAG detection accuracy of at least 95% (DI-5). The team is thinking of employing a pH indicator and a sensing system to alert the medical staff of required maintenance and replacement of treatment systems. These considerations are part of design inputs which require an indicator of WAG concentration (DI-6) and a visual indicator for the alert/alarm (DI-16). As such, the team must consider cost, material availability, ease of use, effectiveness, and reproducibility for any created prototype and final design.

Another crucial external factor is safety, which is of the utmost importance in the OR. Ensuring the pressure in the breathing circuit is not excessive, and any sort of active scavenging is downstream from the breathing circuit will help prevent any catastrophic damage to the patient's lungs, another user need. Maximum negative pressure and pressure drops are therefore critical design inputs in the pathway that ought to be evaluated. The blower/fan must be chosen based on pressure-flow, not just its air flow rate, and its maximum suction to ensure patient safety and effectiveness. The required scavenging flow provided by stakeholders from KidsOR and Public Invention is around 4 L/min. Safety

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for medical staff in the OR is important, as well as individuals outside of the OR. The current method of removal and passive system introduce many hazards. The current tubing is routed on the floor, so it may be stepped on, kinked, damaged, a tripping hazard, or even disconnected. These all can reduce gas flow. Tubing geometry and material are also important safety factors. Friction and inconsistencies in pressure due to geometry could affect the effectiveness of gas removal. The current tubing is weak, is a tripping hazard, and is constantly under stress from being kinked or stepped on. The team is considering smoother alternatives, like PVC, and plans to compare pressure drop and flow between the smoother alternative and the corrugated tubing. Tubing diameter, length, and fittings are all influential for the pressure required and support the user needs of adequate WAG airflow. The design inputs would include the scavenging flow rate, pressure drop, and maximum allowable tubing length whilst maintaining adequate pressure and flow.

Welfare is another important external factor, as quality of care for patients should be maintained or improved without creating significant disturbance in facility workflow. For facilities in lower-resource settings, disruptions for facilities with already limited resources can greatly decrease quality of care, stress out medical staff, and cause the design to not be used. The system should not interfere with the anesthesia delivery or normal OR workflow and should be compact enough to function in various room layouts. This supports UN-4 and UN-8, user needs which require a versatile, easily transportable, and easily manageable device. The Design Control Matrix includes requirements for a safe gripping point or handle (DI-4) as well as a device mass below 25 pounds for portability (DI-8). The design should be simple to operate, maintain, and repair so the personnel on site can continue usage even after KidsOR's three year in-person assistance period. Simple operation and maintenance is shown in the requirement that the device can be disassembled or assembled in less than 10 minutes (DI-9). Additionally, this supports the user need for a service life greater than 5 years (DI-7). Given access to parts and materials may be limited, the team should consider material cost, availability, and ease of replacement when choosing materials for the solution such as tubing, gas treatment, and sensors. The open-source and low-cost need (UN-12) are supported by DI-12 and DI-13, which discuss the need for accessible and reproducible methods and components as well as inexpensive materials.

Another important part of safety is regulatory requirements. Utilizing the FDA Product Classification database, gas scavenging apparatuses have the 'CBN' product code, regulation number 21 CFR 868.5430, and are listed as Class II medical devices with a 510(k) premarket notification pathway. Two relevant medical devices are the Dynamic Gas Scavenging System 2 (DGSS-2), K193646, and the MINISCAV K160950 [4]. The DGSS-2 scavenges WAGs from anesthesia machines during general anesthesia for adults and children. This system is placed between the anesthesia machine and the WAG evacuation vacuum system, utilizing pressure sensing and flow control to prevent excessive pressure from reaching the patient's breathing circuit [5]. This device illustrates a great example of a safe patient, anesthesia machine, and scavenging interface. The MINISCAV is also a Class II gas-scavenging device. It is a small, portable vacuum pump which provides a source for WAG evacuation [6]. The MINISCAV’s intended use is for nitrous oxide, not the anesthetics used in general anesthesia or the KidsOR gases. However, the design is certainly useful and relevant to our final solution as it is meant to be a standalone active system and its testing could be relevant to our benchtop testing and COMSOL modeling. During prototype development, testing practices such as securing tubing and electrical components, checking for leaks, using appropriate PPE, and having appropriate machine training will be observed. The preliminary testing would be benchtop instead of being performed on a

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patient. Both examples illustrate that the relevant regulatory pathway for our solution is a Class II, 510(k) pathway.

Ethical and Professional Responsibilities

Code of Ethics and Study Approval Committees

From the biomedical engineering code of ethics, the team plans to implement the subsections of 2.1.1, 2.1.3, 2.1.4, 2.2.2, 3.1.1, and 4.2.1 [7]. The start of our design process was identifying the problem. Based on the 2.1.1 standard for research and development, the team created a problem statement for a solution that helps hospitals in low-resource areas create an AGSS. By doing this, doctors and OR personnel will not be affected by the symptoms of WAG exposure, and procedures will be done safely. We also spoke with stakeholders from Public Invention and KidsOR that gave us an in-depth understanding of the current issue and how they can support throughout the entire design process. Based on the 4.2.1 standard for mentee responsibilities, the team has created weekly meetings with our department advisor on the project, Dr. Stone, and biweekly meetings with the main stakeholders of the project. By doing this, we are ensuring the team stays aligned with our problem statement and receiving frequent commentary from our clients. We also interviewed external stakeholders. For example, an anesthesiologist, pediatrician and pulmonologist, director of a free clinic, and BMET. Based on the 2.1.4 standard for research and development, the team will continue to reach out to external stakeholders who are more versed in the space of anesthetic waste gas systems, to verify that system that is being built is usable and the team isn't missing any critical information. Finally, as asked by the team's senior design requirements and by our stakeholders, there will be public access to all the information and iterations of our design for others to use. This falls under standards of 2.1.3, 2.2.2, and 3.1.1. By storing all our documents on GitHub, an open-source repository, other people can use the team's findings and create their own system. One of our main goals is to make sure that after the solution has been created, other hospitals, outside of KidsOR, can also implement their own standalone scavenging system. By doing this, low and middle resource hospitals can improve how their medical OR team safely performs procedures.

In the fall, the team plans to complete a preliminary risk analysis to identify potential safety concerns and develop a solution that would reduce the chances of these concerns occurring. In DH02, the team completed a Risk and Hazards Canvas (Figure 1) to evaluate, and numerically rank risk based on severity and probability. A rank of severity from 1-5 would show how serious the danger could be, and a rank of probability from 1-5 would show how likely that danger is to occur. By looking at these two analyses together, the team can create a solution that minimizes harm to people or the environment, while also creating a solution that prevents people in the OR from being affected by WAG. In the spring, the team plans to do failure modes and effects analysis. Once the solution has been created, the team can analyze what could go wrong or fail. With this information and depending on the time constraint, the team can further optimize the solution. Due to the open-source nature of the project, it is also important to highlight and document where the solution could be improved or cause risk, to support other people to add their own iterations onto the project.

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For the fall, the team has reached out to Etta Hanlon, a Virginia Tech Carilion School of Medicine student. She is currently researching how anesthetics gases like desflurane and nitrous oxide contribute to climate change and the carbon footprint of ORs [8]. The team has sent her a list of questions and how she tests these gases. The goal is to gain a further understanding of desflurane and nitrous oxide, and to potentially reach out to Neil Macdonald, M.D., the chair of the anesthesiology department at Carilion Clinic. This way the team can potentially gain access to these gases, and test how they react using our planned solutions. For example, using lye or cat litter to denature the gases. With this information, the team can then finalize the design draft and start ordering parts in the spring. By the end of the spring semester, the solution should be finished and tested. For testing, the team will need access to the gases again. In both the fall and spring semester, depending on the way the team gets access to the gases, there will be training and signed documents in order to use and test these gases. It is also important to remember while testing these gases that they are greenhouse gases and need to be tested in ways that will not be more problematic for the environment.

Ethical Dimensions and Analysis

Globally, the design needs to accommodate differences in WAG exposure limits and hospital requirements across countries. The solution needs to be compatible with different anesthesia equipment, connection ports, tubing, and infrastructure available in low-resource areas. The materials for the solution need to be accessible to low-resource countries and relatively inexpensive. It also needs to be able to be supported by KidsOR for their maintenance and spare part deliveries to the ORs. The tradeoff is creating something that can accommodate all requirements and equipment versus keeping the design cost low.

Culturally, the scavenging system needs to be able to accommodate different anesthetic gases being used by ORs. For example, KidsOR uses isoflurane, desflurane, sevoflurane, etc. The instructions and trainings for this system also need to be easy to understand for medical personnel to use the solution. It is important to keep in mind the countries that are using the solution may speak a language other than English. By making clear, understandable, and easily translatable instructions, the solution can be more widely used. The tradeoff is fitting a system into the original job map versus optimizing the way WAG is removed.

Socially, making a solution that does not require a lot of maintenance or troubleshooting will reduce the load on the BMETs and maintenance workers. The system should also be easy to implement into regular health care professionals' daily lives to make it more usable. With this, the solution would need to provide data that shows it is worth the implementation into the ORs. The tradeoff is minimizing maintenance time versus completing necessary inspections, calibrations, or replacements.

Economically, these machines will be placed in low-recourse ORs and need to be low cost when it comes to buying and installing parts. The maintenance and spare parts of the solution also need to be low cost to make sure the solution continues to work in the hospitals long term. The tradeoff is lowering the price of the solution versus keeping the quality of the solution.

Finally, environmentally, using an active scavenging system allows for direct filtration of WAG. Once filtered, the rest goes back into the operating room and the environmental impacts from the greenhouse gases are reduced. The AGSS must minimize the amount of WAG going into the environment. There

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also needs to be a proper procedure for how to dispose of the gas or denatured gas into liquid once the storage area has been filled.

An environmental ethical tradeoff found in this project is balancing the effectiveness of the AWSS with cost, energy consumption, and material waste involved. One thing the team wants to prevent is creating more medical waste. The global healthcare sector generates 4.5-5.2% of total worldwide greenhouse gases [9]. On top of that, 75.8% of medical waste found in Sub-Saharan Africa is poorly segregated, leading to issues like open dumping or uncontrolled burning of medical and electronic waste that happens in 57.1% of resource-limited regions [10]. When creating medical devices for a low-resource area, it is important to understand that the solution needs to be usable, and the parts need to be accessible. If the team promotes a product that doesn't work or is unusable for doctors and BMETs, the device and all its parts will be wasted. The solution is also meant to reduce the amount of greenhouse gases from the WAG from entering the environment. If the solution doesn't work properly and is hard to use long term, not only would the team be creating more medical waste, but they would also not be solving the greenhouse gas issue either. Overall, the system should be able to help over 100 ORs and prevent OR personnel from falling asleep or getting headaches throughout the day. By additionally tackling the environmental implications of changing from a passive to active scavenging system, the team needs to understand that preventing medical waste, especially because the solution may have electronics, is something that may need to be discussed further when talking about tradeoffs. The passive system that is currently being used relies on a big tubing running across the floor and going out of a window or door. The tradeoff here is that by opening a door or a window, the OR's sanitization is compromised. This ultimately renders the initial solution useless, and the tube is removed entirely. Thus, the WAG is circulating and building in the OR as the procedures occur. By fixing this initial system, it is a fairly simple design; however, it doesn't address the environmental impacts of the WAG that would be pushed directly outside. An active system would allow for the filtration of the WAG, reducing what is entering the air inside and outside of the OR. This system is difficult to design, and the chances of lower usability are higher. Overall, the team has decided to tackle the standalone AGSS in hopes to improve both issues in the OR and environment coming from WAG. This system is technically complicated and involves a lot of chemistry. The team is excited to learn more about this and will keep in mind the potential issues with usability. By creating bi-weekly meetings with main stakeholders, the team hopes that a solution will be created that can help everyone involved and be used.

Product Requirements

Design Inputs

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Table 002-3 Design Inputs

No.	User Need	No.	Design Input	Values and Units	No.	Design Output	No.	Verification	No.	Validation	Priority	Rationale	Source
Performance and Physical Requirements													
UN-1	A device that can be independently powered.	DI-1	The device shall contain a backup power source.	>= battery (voltage TBD)	DO-1		VER-1		VAL-1		3	The OR itself uses a lot of power for the machines already there. Most of these countries that host the OR do not have the money or means to use more electricity.	Dr. Robert Read
UN-2	A device that can run on independent power for an extended amount of time.	DI-2	The device shall be able to run on an independent power source for a couple of hours.	> 10 hours							3	The ORs that will use this product are based in Africa, and there are regular power outages that extend 10 hours or longer before restoration.	Dr. Andy Muelenaer
UN-3	A device with a method to actively remove WAG from the OR.	DI-3	The device shall remove most WAG from the OR.	>= 90% WAG concentration reduction	DO-3		VER-3		VAL-3		5	Our device must remove most of the WAG from the OR for the medical personnel to operate safely.	Courtney Ludick
UN-4	A device that can be transported easily.	DI-4	The device shall have a method to safely grab the container.	>= 1 handle	DO-5		VER-5		VAL-5		3	The OR space is small and has the potential for many tripping hazards, so the device must be easy to move out of the way.	Courtney Ludick
UN-5	A device that can accurately detect WAG concentration in the OR.	DI-5	The device shall have a method to monitor the WAG concentration in the filter.	>= 95% detection accuracy							5	The main purpose of our device is to lower gas concentrations in the OR, so we must monitor the gas concentration in the OR to ensure the	NFPA 99

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												device is working properly.	
UN-6	A device with an output display that is easy to read.	DI-6	The device must have a method to alert when WAG concentration is too high/when to change the filter.	Red indicator for high concentration; Yellow indicator for medium concentration; Green for low concentration	DO-6		VER-6		VAL-6		5	Anesthesiologists must focus on ensuring the patient is safe and needs to be able to quickly view if the waste gas levels are too high in seconds.	Dr. Robert Read/ IEC 60601-1-8
UN-7	A device with an average life cycle.	DI-7	The device shall have a total life cycle that aligns with the life cycle of class 1 medical devices.	>= 5 years							3	The device should not need to be completely replaced for a few years, so the cost for community hospitals can be lowered.	ISO 13485:2016
Human Factors													
UN-8	A device that can be handled safely by all intended users.	DI-8	The device shall have a total mass that permits safe operation.	< 25 lbs.							2	Any person should be able to pack up the device, set up the device, and/or read the device in case the technicians are not available.	Dr. Robert Read
UN-9	A device that can be assembled and disassembled easily.	DI-9	The device shall have an efficient assemble/disassemble time.	< 10 mins							2	Our client performs around 10 surgeries per day, so they need to set up equipment quickly to start on time.	Dr. Robert Read
Environmental Requirements													
UN-10	A device that doesn't push anesthetic gas into the outside air.	DI-10	The device shall have a method of waste gas collection.	< 100 mL/min of gas leakage/operation.							4	Anesthesiologists are looking for ways to reduce the amount of WAG from entering the Earth's atmosphere.	Dr. Brian Struyk
UN-11	A device that can operate in harsh climates.	DI-11	The device shall withstand high temperatures and high humidity.	Can operate between 70-95 degrees Celsius and 80% humidity.							4	The climate where these ORs are located is harsh, often with high temperature and high humidity with little ventilation, so	Dr. Robert Read

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											the device must be adaptable to that climate.	
Manufacturing Requirements												
UN-12	A system that is open-source.	DI-12	The device shall contain product/methods that are accessible and reproducible.	> 90% reproducibility						5	What is being made must be open and accessible to the public to build, add to, and comment on.	Dr. Robert Read
	A device that is low cost.	DI-13	The device shall have inexpensive materials.	< $100						5	The device is being created for a nonprofit organization and is for low-income hospitals, so it should be inexpensive to limit costs the hospital is paying.	Dr. Robert Read
Sterilization, Labeling, Packaging Requirements												
UN-13	A device that can meet the OR's machine sterilization requirements.	DI-14	The device shall have material that can be wiped with a sterilization cleaning solution without affecting performance.	Does the cleaning solution rust or warp the material: yes or no						4	OR machines need to meet the hospital's own sterilization and machine cleaning standards. This way the device can be easily implemented into the typical OR equipment and machine cleaning procedures.	Dr. Brian Struyk
UN-14	A device that can be cleaned regularly	DI-15	The device shall have parts that attach and detach easily for proper cleaning access.	>= 99% of the parts must be accessible to be cleaned						3	The device needs to be easy to clean to prevent risking OR sterilization because of hard to reach or incense cleaning schedules and maintenance.	Dr. Brian Struyk
UN-15	An alarm system that is clearly labeled	DI-16	The device shall have a visual indicator to indicate alarm system/alarm reset.	>= 1 red tape indicator						4	United States regulations require clearly labeled alarm systems, and even though other countries will also be using this system, the premise of the	IEC 60601-1-8

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												standard should be followed as a need.	

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Preliminary Hazard Analysis (PHA)

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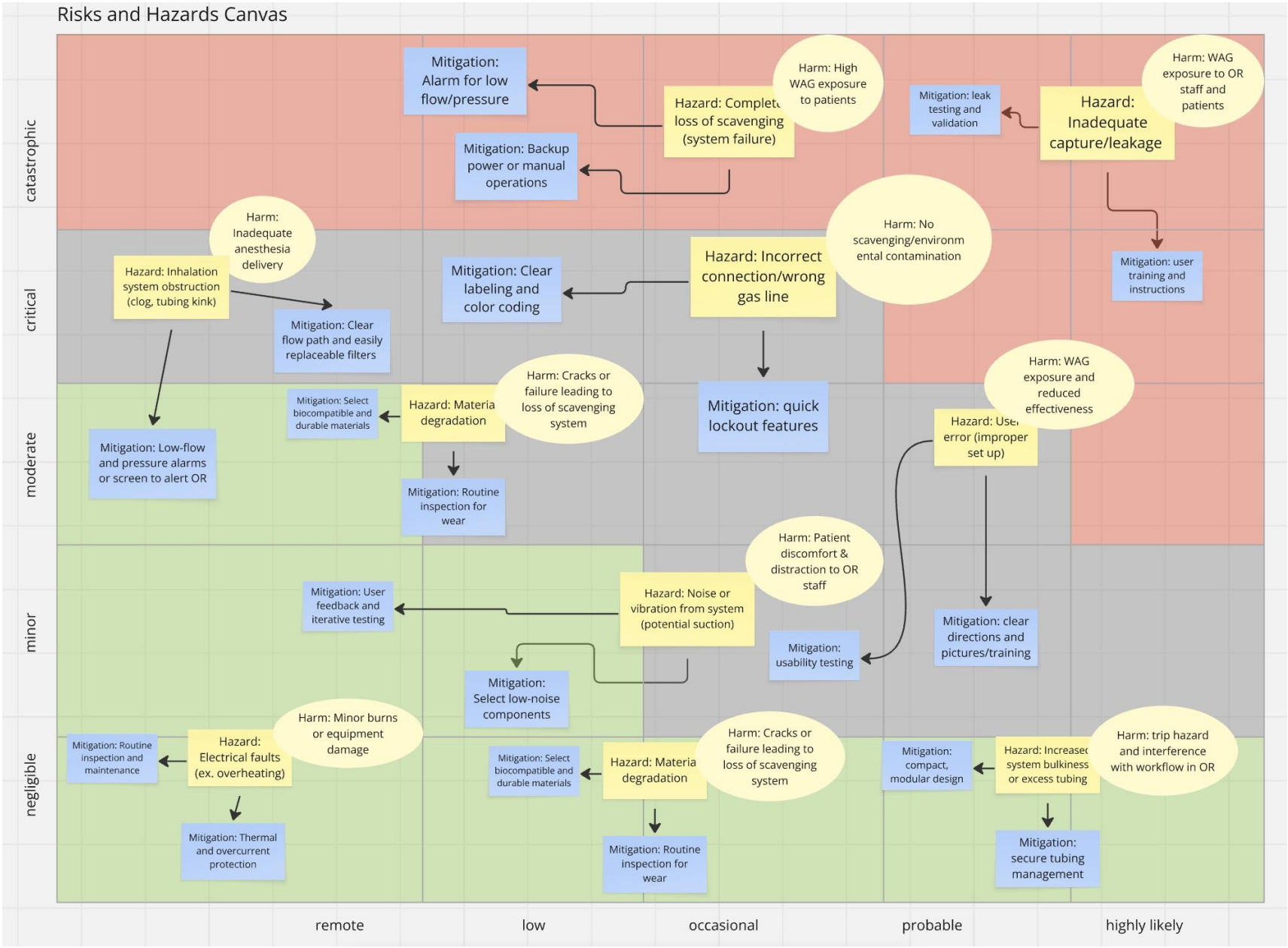
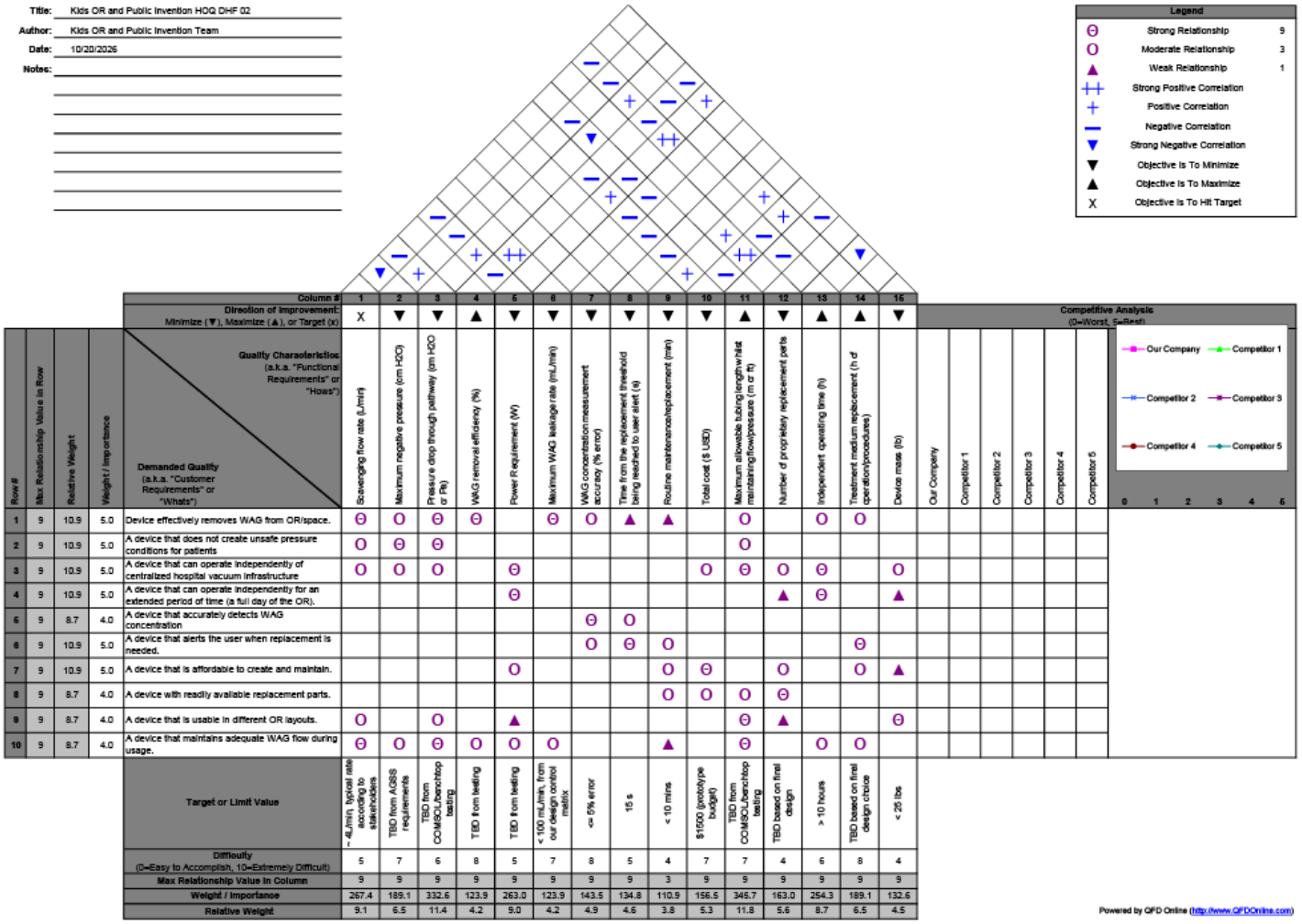


Figure 1: Preliminary Hazard Analysis (PHA) using Risks and Hazards Canvas

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References

[1] O. for, “ISO 7396-2:2007,” ISO, 2025. https://www.iso.org/standard/41945.html (accessed Oct. 20, 2026).
[2] “The Environmental Impact of Inhaled Anesthetics,” ASALifeline. Accessed: Oct. 20, 2026. [Online]. Available: https://www.asahq.org/about-asa/governance-and-committees/asa-committees/environmental-sustainability/greening-the-operating-room/inhaled-anesthetics
[3] T. T. Mnisi, T. Bongongo, and A. M. Nkoane, “Patients’ Views of the Healthcare Services During Power Outages in Pretoria Primary Healthcare,” *Journal of the Colleges of Medicine of South Africa*, vol. 4, no. 1, Jan. 2026, doi: 10.4102/jcmsa.v4i1.254.
[4] FDA, “510(K) Premarket Notification,” Fda.gov. Accessed: Oct. 20, 2026. [Online]. Available: https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfPMN/pmn.cfm
[5] Anesthetic Gas Reclamation, Inc., “Substantial Equivalence Determination for Dynamic Gas Scavenging System 2 (DGSS-2).” Food and Drug Administration (FDA), Nov. 06, 2020. Accessed: Oct. 20, 2026. [Online]. Available: https://www.accessdata.fda.gov/cdrh_docs/pdf19/K193646.pdf
[6] U.S. Food and Drug Administration, “Substantial Equivalence Determination for MINISCAV™ Gas-Scavenging Apparatus,” U.S. Food and Drug Administration, Silver Spring, MD, May 2017. Accessed: Oct. 20, 2026. [Online]. Available: https://www.accessdata.fda.gov/cdrh_docs/pdf16/K160950.pdf
[7] Z. Danziger, “BMES Code of Ethics,” 2021. Accessed: Oct. 20, 2026. [Online]. Available: https://www.bmes.org/hubfs/Files/BMES_Code_of_Ethics_2021_-_Revised_October_2021.pdf
[8] “Greener Anesthesia Cuts Operating Room Emissions in Half Slashes Anesthetic Costs Nearly 40%,” ASALifeline. Accessed: Oct. 20, 2026. [Online]. Available: https://www.asahq.org/about-asa/newsroom/news-releases/2026/10/greener-anesthesia-cuts-operating-room-emissions-in-half-slashes-anesthetic-costs-nearly-40-percent
[9] Y. Wei, M. Cui, Z. Ye, and Q. Guo, “Environmental Challenges From the Increasing Medical Waste Since SARS Outbreak,” *Journal of Cleaner Production*, vol. 291, p. 125246, Apr. 2021, doi: 10.1016/j.jclepro.2020.125246.
[10] A. W. Woime, “Medical Waste Disposal Practices and Their Health Consequences in Resource-Limited Settings: A Systematic Review and Meta-Analysis,” *BMC Public Health*, vol. 26, no. 1, Feb. 2026, doi: 10.1186/s12889-026-26722-9

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